Adaptive Quantum Cryptographic Protocol: Robust Deep Reinforcement Learning Architectures for Real-Time Parameter Optimization

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Abstract. The Quantum key Distribution technique is almost good for real communication environmentse. when it comes to real time communicating through optical channels are sometimes unstable. This is due to two major issues like Atmospheric turbulence and channel noises, which may affect the communication. In particular, These situations may dynamically change overtime which affects the secure communication. Generally, QKD systems use protocols like BB84 protocols which usually use fixed (static) parameter settings. this research proposed that usage of machine learning for making the QKD adaptive. where it overcomes the problem of assuming the parameters are static. In this research, the idea is to adjust parameters automatically which is learn from the communication channels. With that reinforcement learning can be used with a Deep Q Network. To prevent the overestimation problems in RL we use a dual network architecture. Where the post processing parameters such as Privacy amplification ratio and Authentication tag length is modified by Rl agent. In this research the binary entropy of the error rate is used to calculate the learning reward. In this process the error rate measured is called Quantum Bit Error Rate. For secure key generation the system uses security according to Devetak winter bound. On a comparsion to static systems, the proposed system's results were a 79.66% increase in total secure key generation and a 55.89% reduction in block failure rate. These results proves that reinforcement learning can be practically used in improving the robustnes of OKD systems for both dynamic and heterogenous environments .

Keywords: Quantum Key Distribution, Context-Aware Deep Q-Network, Privacy Amplification, Authentication Tag, Quantum Bit Error Rate, Secret Key Rate Optimization

1 Introduction

The security systems in today's internet are started to look fragile because quantum computers are improving every year. There are few modern encryption techniques but they depend on some math assumptions. They are assumed to be hard. This gives root to quantum computing which breaks the assumption. Since it doesnt just increase computational difficulty to protect data but it relies on laws of physics. That is why this make it a better solution for the problem and extremely secure.

To build a QKD systems are not a quite task because these communication channels are continously affected due to several environmental factors like Atmospheric turbulence, beam wandering, stray particles and background noise these all will create a highly unpredictable communication platform. The protocols parameters remain static during these conditions. It means it will fail to adapt fast enough. This leads to bad performance or failure.

The idea of embedding Artificial Intelligence into "cognitive" quantum nodes is the recent research that is being explored. The ability of the systems to observe and adjust the channel on the fly is the goal. For these tasks, Deep Reinforcement Learning (DRL) is is well suited. This paper have revealed serious stability issues during early implementations. The learning process becomes unstable and most of the time, the models fail to converge. This is due to the variable nature of quantum channels.

This paper's work focuses those shortcomings, this process focuses on fixing the engineering problems. Creating another adaptive strategy is not adivsable. The basic principle is to stabilize , context-aware DRL frameworks to perform autonomously for real time conditions without the risk of failures.

1.1 Application and Significance

The deployment of adaptive QKD is successful for the 6G telecommunications technology, where the classiscal and quantum resources are integrating seamlessly through cognitive radio networks.

1.2 Applications of Context Aware Reinforcement Learning

In Quantum Secured IoT, Billions of resource constrained devices are being comprised in Internet of Things(IoT). The available static QKD are too computationally expensive. They continously key hungry for edge nodes. So, for resource allocation, balancing security levels with energy consumption a AI driven approach allows a dynamic solution.

In Satellite Quantum Networks, The connection between the earth to satellite links are expensive. They continously experience severe rapid fluctuations in channel quality. In the study, the adaptive capabilities maximizes the contact windows

during communication.

In some parallel Optimization Domains, The Direct parallels in Quantum image processing have been refined due to iterative optimization techniques. In this application, For quantum image retrieval and reconstruction, a Reinforcement Learning is employed to enhance its quality.

2 Background Study

The development of next-generation secure communication relies on integrating quantum-enhanced algorithms with robust machine learning frameworks. A detailed synthesis of existing literature and the technological roles of prior studies is provided in Table 3.1.

Table 3.1. Structural Overview and Functional Roles of the DRL-QKD Framework

S No	Author Name	Journal & Year	Title of the paper	Observations	Technology used
1	Syed Junaid Nawaz et al.	IEEE Access, 2019	Quantum Machine Learning for 6G Communication Networks: State-of-the-Art and Vision for the Future	In this paper, a framework for 6G networks driven by on demand and self reconfiguration to handle connection problems and sove complex network states. Besides this, this paper also review the State of the art in machine Learning(Ml), quantum Computing(QC), and Quantum ML(QML)	Quantum Machine Learning (QML), Quantum Computing (QC), Deep Learning (DL), QC-assisted ML.
2	Tariq M. Khan and Antonio Robles-Kelly	IEEE Access, 2016	Machine Learning: Quantum Vs Classical	This paper solely discusses the theoretical concepts offered by quantum algorithms and review hardware/software challenges. It also provides a overview of QML which to compared to other classical approaches. These techniques proposed in paper are	Quantum Support Vector Machine (QSVM), Quantum Neural Networks (QNN), Quantum Annealing, Restricted Boltzmann Machines (RBM).

				the potential of quantum systems to process information.	
3	Balasubramani Subbiyan et al.	IEEE Access, 2025	A Quantum-Enhanced Artificial Neural Network Model for Efficient Medical Image Compression	In this paper, the author proposes a way for compressing medical images (MRI, CT, X-Ray) by computing a Quantum enhanced Artificial Neural Network. The images size is reduced compared to the normal standard methods while the model improves the PSNR and SSIM metrics.	Quantum-enhanced Artificial Neural Network (QANN), Quantum feature extraction, Parameterized Quantum Circuits (PQC).
4	Harshala Shingne et al.	MethodsX (Elsevier), 2025	Design of an integrated model using deep reinforcement learning and Variational Autoencoders for enhanced quantum security	This paper uses DRL for dynamic parameter adjustment and VAEs for anamoly detection by integrating AI with quantum security to optimize OKD protocols. This achieves high key generation rates and lowers the quantum bit error rates(QBER)	Deep Reinforcement Learning (DRL), Variational Autoencoder (VAE), Multi-Agent Deep Q-Networks (MADQN), Quantum Key Distribution (QKD).
5	Lalitha Kumari Pappala et al.	IEEE Access, 2025	Design of an Iterative Model for Incremental Enhancements in Quantum Image Processing Using Reinforcement Learning-Based Optimizations	The paper helps in enhancing biomedical image processing by introducing four quantum methods (Q-SVM-EFE, QGAN-DA, QECA-FD, QIR-ND). These techniques have observed and improvements where shown in classification accuracy, synthetic data generation, anaomaly detection sensitivity, and reconstruction speed.	Quantum Support Vector Machines (Q-SVM), Quantum Generative Adversarial Networks (QGAN), Quantum Entanglement and Correlation Analysis (QECA), Quantum Image Reconstruction (QIR).
6	P. Deepanramkumar and A. Helen Sharmila	IEEE Access, 2024	AI-Enhanced Quantum-Secured IoT Communication	This paper employs AI models for efficient spectrum sensing and Channel State	Quantum Key Distribution (QKD), Reinforcement

			Framework for 6G Cognitive Radio Networks	Information (CSI) predication. Where it uses a framework for 6G Cognitive Radio Networks is proposed, featuring dual layer authenticaion (QKD-PKL). This is used to ensure secure and reliable IoT connectivity.	Learning-based Ensemble Regression (RL-ER), Multi-Layer Perceptron with Kalman Filter (MLP-KF), Public Key Infrastructure (PKI).
7	Ying-Yi Hong and Dylan Josh Domingo Lopez	IEEE Access, 2025	A Review on Quantum Machine Learning in Applied Systems and Engineering	This paper categorizes the studies by learning modalities and architectures and it gives a topical review analyzes QML applications across engineering domains like (healthcare, energy, finance, etc). the NISQ era is highlighted by the utility of hybrid quantum classical models.	Quantum Machine Learning (QML), Quantum Deep Learning (QDL), Hybrid Quantum-Classical Models, Variational Quantum Circuits (VQC), Quantum Kernel Methods.

3 Research Gaps

Achieving reliable quantum communication requires overcoming several technical hurdles that current methodologies fail to address effectively. These research gaps are categorized in Table 4.1.

Table 4.1. Identification of Research Gaps in Intelligent QKD Systems

Research Gap	Technical Description	Impact on System
Static Mobility Modeling	Secondary users (SUs) are often assumed for their static mobility by the Exisiting cognitive Radio networks (CRN). The ignore the dynamic movements and changes in signal fluctuations.	Received Signal Strength Indicator (RSSI) has been reduced significantly and there is a significant data loss, which gives compromising network reliability.
Vulnerability in Random Key Generation	Random key generators are used for Conventional Elliptic Curve Cryptography (ECC) which can be used for computationally predictable or exploited by sophisticated attacks.	The IoT data transmissions which uses cryptographic keys creates vulnerable and potential security issues. This makes the systems prone to data breaches.

Quantum Repeater Scalability	The process of simple copying of quantum states (qubits) are prevented by the “no cloning theorem”. Which makes it difficult for repeaters in signal amplification, construction fo quantum switches and routers.	The development of long distance quantum communication link and the scalability of quantum internet in 6G networks are made very difficult to implement.
Classical ML Latency	Standard CPUs/GPUs suffer from significant processing delays when handling the massive, multi dimensional data from 6G. This are when running conventional Machine Learning algorithms.	The Ultra Reliable Low Latency Communications (URLLC) are done flawlessly by preventing the real time network orchestration and intelligent management, which is critical.
Inefficient Spectrum Allocation	When dynamically adapting to spectrum usage, the Current allocation strategies often fail, this leads to underutilization the signification portions of remaining spectrum.	It worsens the issue of 6G cognitive IoT networks by spectrum scarcity, also limiting the quantity of supported devices. This also worsens the system by reducing the throughput.
Diagnostic Quality Loss	When compressing high resolution medical data images, tradional compression standards (e.g., JPEG, PNG) often discardes the fine details and introduce artifacts.	Affects the telemedicine and remote diagnosis by compromising the diagnositic accuracy of medical imaging systems (MRI, CT, X-ray)
Centralized Control Complexity	Centralized management becomes computationally prohibitive and creates single point of failure during the time when quantum networks scales in size.	When implementing Decentrailized solutions like Multi agent Deep Q Networks (MADQN) to maintain efficiency, it increases the computational overhead and vulnerability.
NISQ Hardware Constraints	The High noise levels, limited qubit counts and short coherence times significant affect the Current Noisy Intermediate Scale Quantum devices.	Necessiates the use of hybrid quantum classical architectures. it restricts the complexity of Quantum machine Learning(QML) models the can be fully deployed.

4 Problem Statement

A Binary failure mode is faced by the Static Quantum Key Distribution (SQKD) where the operating systems are prone to turbulent weather. When tuning the parameters it may be too conseravative during clear weather inorder to withold the aggressive weather and turbulence. This results in wasting capacity. Leading the

Security breaches and aborting of strict protocols. when implementing “naive” Reinforcement learning (RL), it suffers from more software level instabilites where it is prone to replay buffers sequencing errors. Either that or it diverges due to “moving target problem”, which is inherent in Q-Learning. A three layer architecture has been given to address this limitation. This three layer architecture of the Safe-DRL QKD is shown in the Fig 4.1.. A physical system for environment monitoring is decoupled from the system. For predicitve adaptation an Intelligent control layer is used by the neural network policy. Additionly a dedicated Safety and Training layer is used. By intercepting unsafe actions through predefined rules and limits, it is ensured that the computational stability and security compliance are compromised. This results in mitigating risks while maintaining optimal key rates before they reach the hardware actuator.

5 Materials and Methods

For the QKD control plance, we specifically engineered and introduced a Context-Aware Deep Q-Network (DQN)

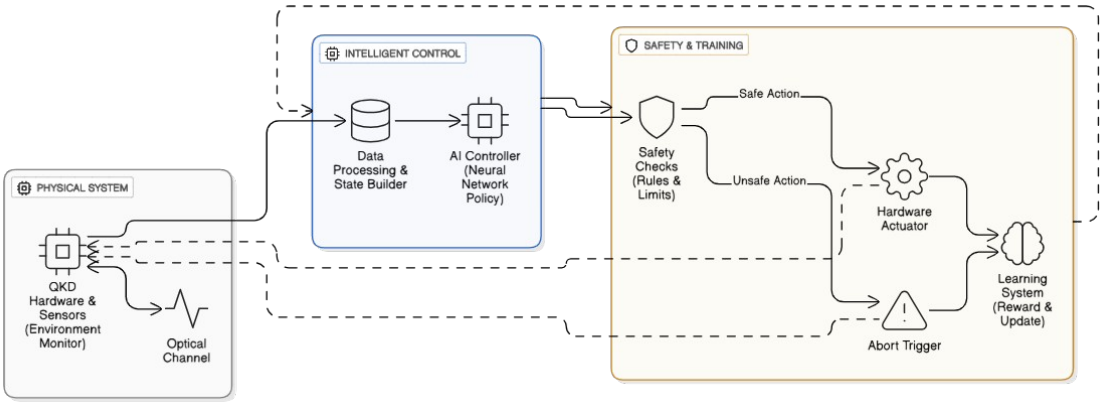

Fig 5.1. Proposed Three-Layer Architecture for Safe-DRL Optimized Quantum Key Distribution. S_t

5.1 Context-Aware State Space:

We introduce a new state vector S_t to include a momentum term, shown in (1) since the standard RL agents see only a single snap shot or a single state of the channel. Here QBER represents the derivate of the error rate.

$$S_t = [QBER_t,\ \Delta QBER,\ SNR_t,\ PoolStatus,\ EveIndicator] \ldots\ldots\ldots(1)$$

Due to this, the random noise spikes (zero momentum) and real atmospheric turbulence (positive momentum) are very well distinguished. This enables proactive throttling of transmission rate.

5.2 Entropy-Based Reward Function:

To quantify to maximum information and eavesdropper could possess, the reward function R_t is returned in a way that it is derived from the binary entropy function H_2(p) . The equation (2) which shows both the informatic and theoratical securites are mutually respected by the agent. A ensure that the agent does not cross the threshold

$$X_t = R_{raw} * (1 - H_2(QBER) - f(QBER) . H_2(QBER)) \ldots\ldots\ldots (2)$$

As the 11% percent security threshold is approached by the QBER, function induces a server penalty term to the system. So by doing this the agent is forced to prioritize the link survival to maintain communication rather than foucsing on raw speed during the times of instability. This is done by adopting a risk adverse policy.

5.3 Target Network Stabilization:

A dual network architecture is employed to solve the instability problem faced by the standard Q learning. A Q-value targets are generated by the seperate Target Network (θ') while actions are determined by the Policy Network (θ). For every period, The Target Network's weight are frozen and updated, and not continously. This provides a stable anchor for the learning process. By dampening the oscillations, the channel noises are stable.

6 Experiments and Results

A BB84 protocol is used to model the system over a 50km distance with a attenuation of 0.2dB/km. For modelling the simulation environment and the agent python and PyTorch are used.

Algorithm: Context-Aware DQN Training Loop

1 **Initialize** Policy Network Q and Target Network Q̂ with random weights.
2 **Initialize** Replay Memory D.
3 For episode = 1 to M:
a. Reset the ienvironment to initial, get initial state s_1.
b. For step t= 1 to T:
i. Using ε -greedy policy, select action a_t (exploration vs exploitation).
ii. In QKD simulator, Execute a_t so that to adjust PA ratio, Tag Length.
iii. Observe the reward r_t (Entropy-based) and the next state $s_{|t+1|}$(with ΔQBER).
iv. Store transition for next iteration (S_t, a_t, r_t,$s_{|t+1|}$) in D.
v. Sample a random minibatch of transitions from D.
vi. Transpose Batch: For tensor processing separate state/action components.
vii. Calculate loss L = (r_t γ $max_a^{'}$ Q^($s_{|t+1|}$, a') - Q(S_t, a_t))^2.
viii. Perform gradient descent step on Q.
ix. Every C steps, update Q^ ß Q.

Hyperparameters:

1.**Learning Rate:** 0.0005
2. **Gamma (Discount Factor):** 0.9 (Prioritizing long-term link stability)
3. **Batch Size:** 128
4. **Target Update Frequency:** 1000 steps
5. **Action Space:** 40 discrete actions (Combinations of 4 Tag Lengths and 10 PA Ratios).6.2 Comparative Performance Trends

Results and Discussions:

The agent benchmarked against a static baseline too represent the traditional standard industry protocols (Fixed Tag=64, PA=0.5)., Where the agent was trained for 5000 episodes overtimes to enhance the performance.

6.1. Quantum Channel Simulation

Fig 6.1: QKD Signal Intelligence: Simulation Interface for the BB84 Channel Monitor using Deep Reinforcement Learning.i

6.2 Convergence Analysis and Performance Evaluation

The agent shows a smooth, monotonic decrease in temporal difference error because of the Context Aware DQN with Target Networks which is seen in Fig. 6.2. This is a great improvement because while using naive implementations, it was experiencing a great oscillating loss functions overtime.The learning target has been controlled and stable now by using a target network. This process is successfully which now allows the system to converge at a good rate. When following a optimal policy, this agent which used DQN can converge despite of the noisy data.

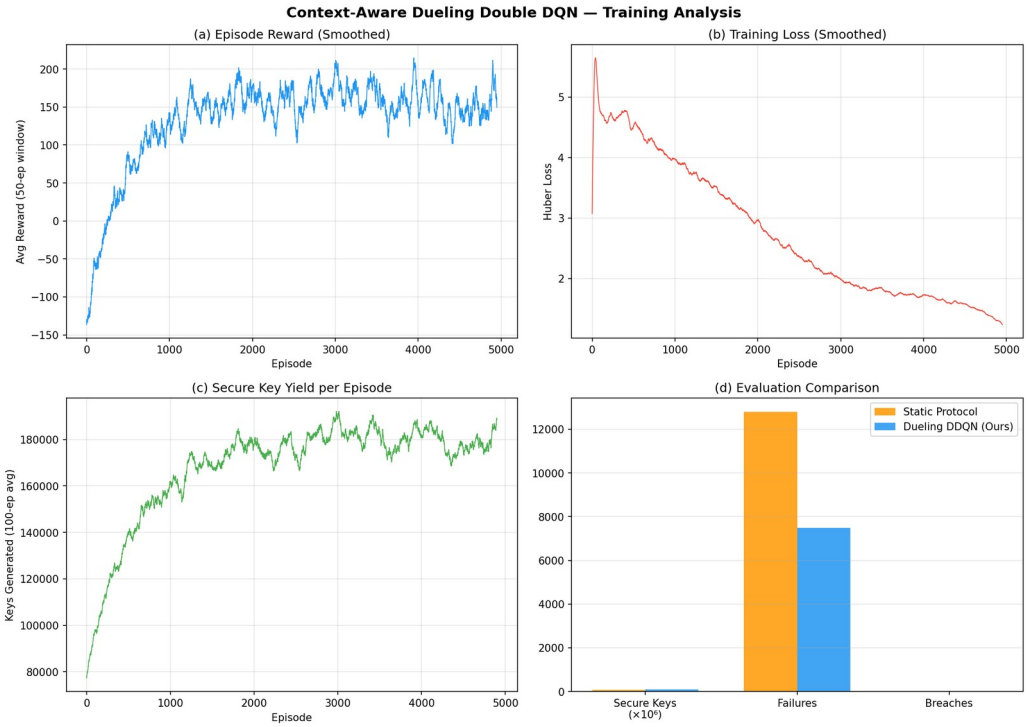

Fig 6.2: Convergence Analysis and Performance Evaluation of the Context-Aware Agent over 5,000 Episodes.

6.3 Performance Metrics

- **Total Secure Keys:** The agent showed a 79.66% increase in throughput, which is calculated using the **122,418,297 secure bits** that are generated by the AI agent, compared to the static baseline's which have only 98,128,634 bits only.

- **Link Stability:** The system shows a reduction of 55.89%. The AI agent failed and aborted the blocks 7,213 times, while the static one numbers 16,352 times.

Which is a great improvement.

6.4 Behavioral Analysis

The "Context-Aware" state space enabled a behavior we term "Pre-emptive Throttling." When the Delta QBER input indicated a rising error trend, the agent would proactively increase the Authentication Tag length and reduce the Privacy Amplification ratio before the QBER breached the security threshold. This anticipated the turbulence spike, keeping the link alive where the static protocol would have aborted connection

A behaviour called “Pre-emptive Throttling” has been enabled by the “Context Aware” state space. The agent is actively increasing and decreasing the Authenication Tag length and The Privacy Amplification ration ratio, when it faces a Delta QBER input that gives a rise in error trend. Before the QBER breaches the security the threshold, the system should manage to perform the certain things. The Static protocol will aborted theses connections that we anticipated due to turbulence spike.

6.5 Comparative Analysis

The performance of the proposed Context Aware DQN against the traditional methods and Rl approaches are described in the Table 6.1.

Table 6.1. Comparative Analysis on Context Aware DQN vs the other approaches.

Metric	**Static Protocol**	**Naive DQN**	**Context-Aware DQN**
Secure Key Yield	Baseline (1.0x)	1.2x (Unstable)	1.79x
Failure Rate	High (>20%)	Moderate (~15%)	Low (~8%)
Adaptation Latency	None (Fixed)	Reactive (High Latency)	Predictive (Low Latency)
Security Model	Fixed Thresholds	Heuristic Reward	Entropy-Based (Devetak-Winter)

7 Conclusion and Future Work

This paper and study concludes that when Deep Reinforcement Learning is engineered with proper and approopriate stabilization means it can perform a high stability control and security aware reward functions which can enhance the robustness of our agent in long time. This makes the process of Quantum Key Distribution to be more robust. Our proposed model outperforms nearly 80% of time compared the the static protocols and it addresses the key limiations of the previous studies and algorithmic instabilities. This proves that the AI is not only a simple auxilary tool but a important component forr the practical deployment for global, weather resistant quantum cryptographic networks.

Multi-Agent Reinforcement Learning (MARL): The idea is to scale the single link agent to a multi node quantum network, to optimize the end to end routing fidelity.

FPGA Deployment: The Trained PyTorch model should not only stay in the local machine itself, rather it should be transfered to FPGA hardwares to reduce the latency in inference from milliseconds to microseconds, which matches the speed of photon detecors.

Continuous Variable QKD (CV-QKD): Adapting the system to use shannon entropy with present entropy based reward function will extend the framework to CV-QKD systems.

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